D 11983 **(Pages : 3)** **Name.....**

Reg. No.....

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION NOVEMBER 2021

Common Course [B.Sc. LRP (Alternate Pattern)]

A11—BASIC NUMERICAL SKILLS

(2019—2020 Admissions)

Time : Two Hours and a Half Maximum : 80 Marks

Section A

Answer at least **ten** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 30.

1. What is power set ?
2. Find the power set of $A = \{1, 2, 3\}$.
3. what are the methods used for measuring seasonal variations ?
4. Represent the following frequency table by histogram :

Marks	:	10–15	15–20	20–25	25–30	30–35
Number of students	:	5	20	50	40	10

5. What is analysis of time series ?
6. Find the product of first five terms of GP, if the third term is 3.
7. What do you understand by classification of data ?
8. Solve $x^2 - 7x + 6 = 0$ by using quadratic formula.
9. Explain Kurtosis.
10. Find the product of first 9 terms of GP, if the 5th term is 2.
11. Find the mean of the following data.

    4, 40, 60, 20, 80, 10, 26, 12, 24, 12, 50

Turn over

12. Explain Skewness.

13. What is a pie diagram ?

14. Find the value of the determinant $\begin{vmatrix} 1 & 0 & 0 \\ 4 & 4 & 2 \\ 2 & 1 & 3 \end{vmatrix}$.

15. What is an index number ?

(10 × 3 = 30 marks)

Section B (Paragraph)

Answer at least **five** *questions.*
Each question carries 6 *marks.*
All questions can be attended.
Overall Ceiling 30.

16. If the fifth and the tenth terms of a G.P are 32 and 1024 respectively, find the first term and the common ratio.

17. Give 3 yearly moving averages for the following series :

Year	:	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Production (lakh tons)	:	12.2	12.3	13.7	14.9	13.2	11.3	15.1	15.2	15.3	14.9

18. Find the sum of first 20 terms of the sequence 3, 6, 9, 12,....

19. Find the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

20. Find the central tendencies for given series :

28, 36, 34, 28, 48, 22, 35, 27, 19,41

21. Find AB, where $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 3 & 1 \end{bmatrix}$.

22. If $A = \{1, 2\}$ and $B = \{a, b, c\}$, find A × B and B × A. Are they equal ?

23. What are the different aspects to be considered in planning a statistical enquiry ?

(5 × 6 = 30 marks)

Section C (Essay)

Answer any **two** *questions.*
Each question carries 10 *marks.*

24. Find the sum of n terms of the series 8 + 88 + 888 + 8888 + ...

25. Find the inverse of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

26. Solve the following by matrix method :

$$2x + 3y + 3z = 5$$
$$x - 2y + z = -4$$
$$3x - y - 2z = 3.$$

27. Find the quartile deviation for the following data :

Marks	Frequency
20–30	4
30–40	12
40–50	18
50–60	28
60–70	19
70–80	14
80–90	5

(2 × 10 = 20 marks)